Adhvaith Vijay (U.S. Citizen)

Product-Focused Data Scientist Turning Models into Decisions
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WORK EXPERIENCE

Tesla | Senior Data Scientist **July 2024 - Present**

- Forecast thermal-runaway risk and warranty life for 10+ vendor and in-house battery cell programs by combining lab stress tests under 98th-percentile fleet conditions with 1 TB of telemetry from 3.5M vehicles; sign off on go/no-go qualification decisions as cell owner
- Own the final quality gate for every newly manufactured Tesla 4680 battery before it enters a vehicle; develop and deploy defect detection models that segment 50+ battery components in CT scans, delivering 98% recall on critical checks
- Saved 200+ technician hours/week by building a quality-analytics pipeline and live dashboards used daily by 45 engineers across 4 teams; the pipeline inspects 2,000+ batteries/day via Celery/Redis on 6 VMs and flags out-of-spec cell metrology through Splunk alerts

Standard AI | Data Scientist **Sept 2023 - June 2024**

- Standard AI uses camera-based 3D shopper tracking to analyze in-store behavior; led product-placement A/B tests for Compass Group USA, randomized by store and week, lifting interaction-to-purchase conversion 24% to 31% (+29%)
- Partnered with product and sales to define shopper engagement metrics (funnels, cohorts, repeat visits) and built an in-house Python library to compute them in Databricks and Snowflake, winning $600K in new contracts over 3 months
- Built a Monte Carlo simulation of shopper paths and shelf-to-shelf travel times to predict how store layout changes shift shopping behavior; recommended product-placement swaps to retail clients that lifted revenue 15% across two stores

Tesla | Cell Engineer Intern **Jan 2023 - Sept 2023**

- Improved defect-detection recall 40% over the previous binary filter with a Vision Transformer model on CT scans, catching core impingement and collapse before cells moved downstream
- Eliminated the need for multiple $27,000 Keyence machines with a CT-based method that non-invasively measures terminal/lid bulge on cells within 50 microns, so engineers inspect cells without new equipment
- Built a daily-refreshing Tableau dashboard linking dissection reports to production metrics, surfacing failure trends that drove electrolyte loading and cell design changes, cutting electrolyte leaks by 20% in 5 months

JPMorgan Chase & Co. | Summer Associate (Credit Fraud Risk) **June 2022 - Aug 2022**

- Enhanced small-business lending decisions by lifting a PySpark loan default model 14% for the Consumer & Community Banking team, engineering domain-specific features for erratic repayment patterns, seasonal revenue shifts, and credit-utilization spikes
- Made high-risk applicant segments explainable to risk stakeholders and federal regulatory auditors by applying feature binning and hierarchical clustering to the lending model's input features, giving clear reasons for each segment

Rolls-Royce North America | Undergraduate Researcher **Oct 2019 - May 2020**

- Built a data pipeline to determine the optimal compressor wash interval for Rolls-Royce AE3007 engines, improving maintenance planning
- Surfaced the pipeline's output in an R Shiny tool with geospatial views of hundreds of at-risk aircraft, cutting the time for maintenance planners to isolate flights needing a compressor wash from 3–4 hours to under 5 minutes

EDUCATION

University of California, San Diego **Sept 2022 - March 2024**
Master of Science in Data Science

University of California, Los Angeles **Sept 2019 - June 2022**
Bachelor of Science, Statistics and Data Science

SKILLS

Languages & ML: Python, R, SQL, scikit-learn, PyTorch, XGBoost
Data & ML Ops Tools: Airflow, Fivetran, dbt, Git, Databricks, Docker, Snowflake, MLflow, BigQuery
Analytics & Visualization: A/B Testing, Tableau, Power BI, Looker, Plotly, Hex
Domain Expertise: Lithium-Ion Battery Reliability, Retail & Consumer Analytics, Product Experimentation, Shopper Behavior Analytics, Consumer Banking Risk

PATENTS

Morales Saiki, L. Y., Vijay, A., Zanlungo, F., Dhingra, A., & Lee, M. J. (2024). *Subject-tracking in a cashier-less shopping store for autonomous checkout for improving item and shelf placement and for performing spatial analytics using spatial data and the subject-tracking* (U.S. Patent Application No. US2024193497A1)

PAPERS

A. M. Vijay, F. Zanlungo, D. M. Woollard, and L. Y. Morales. Analyzing the shopping journey: Computing shelf visits in a physical retail store. In Proceedings of the 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Under review.